SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXPERIMENT 4

Case Study: E-Commerce Application – Positive Test Scenarios and Test Cases

AIM:

To design and execute positive test scenarios and test cases for an E-Commerce Application to verify its functional requirements using valid inputs and ensure that all major functionalities work as expected.

Problem Description

An online shopping application allows customers to register, log in, search for products, add products to a shopping cart, and place orders. Customers must provide valid details during registration and login. The application should validate product availability, quantity, delivery address, and payment details before confirming an order. The testing team is responsible for verifying whether the application works correctly under valid and invalid conditions. They must prepare positive and negative test cases and document any defects identified during testing.

Functional Requirements

1. Users must register with a valid name, email address, password, and mobile number.

2. Users must log in using valid credentials.

3. Products must be added to the cart only when they are available.

4. The requested quantity must be greater than zero and must not exceed available stock.

5. The application must accept valid delivery details.

6. An order must be confirmed only after successful payment.

7. The application must display appropriate error messages for invalid inputs.

2. Positive Test Cases

Positive testing verifies whether the application works as expected when valid inputs are provided.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Register a new user	Valid name, email, password, and mobile number	Account created successfully	Pass
PT02	Login with valid credentials	Registered email and correct password	User logs in successfully	Pass
PT03	Search for a product	Search for “Laptop”	Relevant products displayed	Pass
PT04	Add an available product to cart	Select an in-stock product	Product added to cart	Pass

PT05	Update cart quantity	Change quantity from 1 to 2 when stock is sufficient	Quantity and total price updated correctly	Pass
PT06	Enter valid delivery address	Valid name, address, city, PIN code, and mobile number	Address accepted	Pass
PT07	Make successful payment	Valid payment details and successful response	Payment succeeds and order is confirmed	Pass
PT08	View order history	Login and open order history	Previous orders displayed correctly	Pass

3. Conclusion

The positive test cases verify that the E-Commerce Application performs its required operations correctly when valid inputs and conditions are provided.